



Preparing to solve two step linear equations

Task A: Using representations to solve a two step equation

1) Use the bar models to find a solution for x in each question.

a) <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>x</td><td>x</td><td>x</td><td>x</td><td>3</td></tr> <tr><td colspan="5">35</td></tr> </table> <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>x</td><td>x</td><td>x</td><td>x</td></tr> <tr><td colspan="4">32</td></tr> </table> <table border="1" style="width: 100%; text-align: center;"> <tr><td>x</td><td>x</td><td>x</td><td>x</td></tr> <tr><td>8</td><td>8</td><td>8</td><td>8</td></tr> </table> Solution when x is 8	x	x	x	x	3	35					x	x	x	x	32				x	x	x	x	8	8	8	8	b) <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>x</td><td>x</td><td>22</td></tr> <tr><td colspan="3">72</td></tr> </table> <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>x</td><td>x</td></tr> <tr><td colspan="2">50</td></tr> </table> <table border="1" style="width: 100%; text-align: center;"> <tr><td>x</td><td>x</td></tr> <tr><td>25</td><td>25</td></tr> </table> Solution when x is 25	x	x	22	72			x	x	50		x	x	25	25	c) <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td colspan="6">9</td></tr> <tr><td>4</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> </table> <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td colspan="5">5</td></tr> <tr><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> </table> <table border="1" style="width: 100%; text-align: center;"> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> </table> Solution when x is 1	9						4	x	x	x	x	x	5					x	x	x	x	x	1	1	1	1	1	x	x	x	x	x	d) <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>12</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> <tr><td colspan="6">50</td></tr> </table> <table border="1" style="width: 100%; text-align: center; margin-bottom: 10px;"> <tr><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> <tr><td colspan="5">38</td></tr> </table> <table border="1" style="width: 100%; text-align: center;"> <tr><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td></tr> </table> Solution when x is $7.6\frac{38}{5}$ or 7.6	12	x	x	x	x	x	50						x	x	x	x	x	38					x	x	x	x	x					
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2) Match each bar model to an equation in the form $ax + b = c$

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Task B: Writing linear equations in the form $ax + b = c$

1) Write each equation in the form $ax + b = c$. You do not need to solve the equation.

a) $7x + 3 + x + 1 = 12$
 $(7 + 1)x + (3 + 1) = 12$
 $8x + 4 = 12$

b) $5 - 3x + 5x - 8 = 1$
 $(-3 + 5)x + (5 - 8) = 1$
 $2x - 3 = 1$

c) $10 - x + 7 - 4x = 2$
 $(-1 - 4)x + (10 + 7) = 2$
 $-5x + 17 = 2$

d) $3(x - 2) + 5(x - 2) = 16$
 $8(x - 2) = 16$
 $8x - 16 = 16$

e) $4(x - 5) + 5(3x + 2) = 85$
 $4x - 20 + 15x + 10 = 85$
 $19x - 10 = 85$

f) $6(4 - 2x) - (4 - 2x) = -40$
 $5(4 - 2x) = -40$
 $20 - 10x = -40$

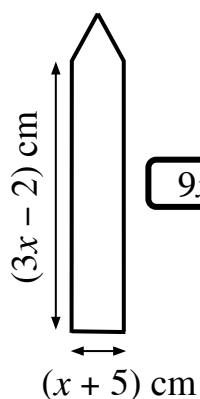
2) Match the shapes below to one of these expressions for its perimeter.

$10x - 4$

$10x - 2$

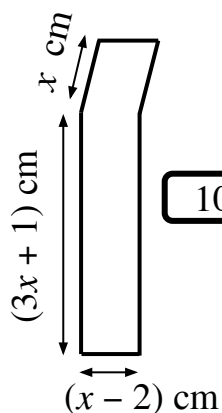
$9x + 11$

a) An equilateral triangle on top of a rectangle



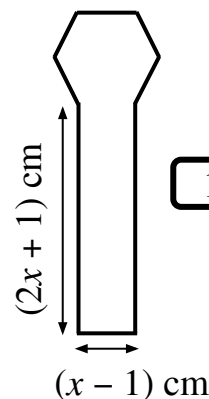
$9x + 11$

b) A parallelogram on top of a rectangle



$10x - 2$

c) A regular hexagon on top of a rectangle



$10x - 4$

